

Solution STA510 Exam fall 2022

Problem 1

a)

For $f(x)$ to be a proper density, we must have that $\int_0^4 f(x)dx = 1$. Here

$$\int_0^4 f(x)dx = (4C)^{-1} \int_0^4 \exp(x/4)dx = C^{-1}[\exp(x/4)]_{x=0}^4 = C^{-1}(\exp(1) - 1).$$

solving $\int_0^4 f(x)dx = 1$ we obtain

$$C^{-1}(\exp(1) - 1) = 1 \Rightarrow C = \exp(1) - 1.$$

which is what was to be shown.

b)

For the mean, we have (see tables and formulas, section 7)

$$E(X) = (4C)^{-1} \int_0^4 x \exp(x/4)dx = (4C)^{-1} 4^2 [\exp(x/4)(x/4 - 1)]_{x=0}^4 = \frac{4}{C} (0 - (-1)) = \frac{4}{C}.$$

Further, to calculate $Var(X) = E(X^2) - E(X)^2$, we need to find $E(X^2)$ which may be expressed as the integral:

$$E(X^2) = (4C)^{-1} \int_0^4 x^2 \exp(x/4)dx$$

For completeness, not required:

$$\begin{aligned} E(X^2) &= (4C)^{-1} \int_0^4 x^2 \exp(x/4)dx = (4C)^{-1} 4 [\exp(x/4)(x^2 - 8x + 32)]_{x=0}^4 \\ &= \frac{1}{C} (\exp(1)(16 - 32 + 32) - (32)) = \frac{16 \exp(1) - 32}{C} \end{aligned}$$

The variance is given by $E(X^2) - (E(X))^2$:

$$\begin{aligned} Var(X) &= \frac{16 \exp(1) - 32}{C} - \frac{16}{C^2} = \frac{C(16 \exp(1) - 32) - 16}{C^2} \\ &= \frac{C(16C - 16) - 16}{C^2} = 16 \frac{C^2 - C - 1}{C^2} = 16(1 - C^{-1} - C^{-2}) \end{aligned}$$

which gives the formula given in the exam.

c)

The cumulative distribution function, for $0 \leq x \leq 4$ obtains as

$$F(x) = \int_0^x f(t)dt = \frac{1}{4C} \int_0^x \exp(t/4)dt = \frac{1}{4C} 4 [\exp(t/4)]_{t=0}^x = C^{-1}(\exp(x/4) - \exp(0)) = C^{-1}(\exp(x/4) - 1)$$

which is what is given in the question. Sanity checks (not really required:

$$F(0) = C^{-1}(\exp(0) - 1) = 0, \text{ OK!}$$

$$F(4) = C^{-1}(\exp(4/4) - 1) = \frac{C}{C} = 1, \text{ OK!}$$

Now calculate the probabilities using the CDF:

$$P(X > 1) = 1 - P(X \leq 1) = 1 - F(1) = 1 - C^{-1}(\exp(1/4) - 1) \approx 0.8347038231$$

$$P(1 < X < 2) = F(2) - F(1) = C^{-1}(\exp(2/4) - \exp(1/4)) \approx 0.2122444922$$

d)

The inverse CDF is given as the solution in x to $F(x) = u$, $0 \leq u \leq 1$:

$$F(x) = C^{-1}(\exp(x/4) - 1) = u \Rightarrow \exp(x/4) = Cu + 1 \Rightarrow x = 4\log(Cu + 1) = F^{-1}(u).$$

Sanity check (not really required): we should have that $F^{-1}(0) = 0$ and $F^{-1}(1) = 4$:

$$F^{-1}(0) = 4\log(1) = 0, \quad F^{-1}(1) = 4\log(C + 1) = 4\log(\exp(1)) = 4, \quad \text{both OK!}$$

Second part: In order to simulate n random variables $x_1, \dots, x_n$ iid with density $f(x)$ one needs to do the two steps

- simulate n iid $\text{uniform}(0,1)$ random variables, say $u_1, \dots, u_n$.
- apply the inverse CDF to each u_i to obtain x_i , i.e. $x_i = F^{-1}(u_i) = 4\log(Cu_i + 1)$, $i = 1, \dots, n$.

e)

In order to do AR-sampling based on $\text{uniform}(0,4)$ proposals, we first must find c so that $f(x)/g(x) = 4f(x) \leq c$ for all $0 \leq x \leq 4$. Now, $f(x)$ is maximized at monotonously increasing, hence $4f(x) \leq 4f(4)$, and therefore we may choose $c = 4f(4) = \exp(1)/C$.

This is indeed the c chosen in lines 5 and 13 in the code of the exam.

Looking further at the former function `p1.ar.1` we see that proposals are made according the proposal distribution $g(x)$ (line 7), and that proposals are accepted with probability $f(x)/(cg(x))$ (line 8). We may then conclude that `p1.ar.1` produces correct samples.

Looking at `p1.ar.2`, we see that proposals are made in line 15 using `4.0*runif(1)`. This is still a draw from the $\text{Uniform}(0,4)$ distribution, as it is just a scaling of a $\text{Uniform}(0,1)$ distribution to the appropriate range (may be shown properly as $F(x) = x/4 \Rightarrow F^{-1}(u) = 4u$). Further we see in line 16 that proposals are accepted with probability $f(x)/c$. The code will still produce correct results as $f(x)/c = f(x)/(4c(1/4)) = f(x)/(c'g(x)) < 1$ where $c' = 4c$. Hence, the code is correct, but it is unnecessarily slow.

f)

Since $\bar{X}$ is an average, the central limit theorem tells us that it will be approximately normally distributed. Then it suffices to find the mean and variance of $\bar{X}$:

$$E(\bar{X}) = n^{-1}E\left(\sum_i X_i\right) = n^{-1}\sum_i E(X_i) = n^{-1}nE(X_1) = E(X_1) = 4/C.$$

I.e. the average has mean equal to the expected value of each of the X_i s.

$$\text{Var}(\bar{X}) = n^{-2}\text{Var}\left(\sum_i X_i\right) = n^{-2}\sum_i \text{Var}(X_i) = n^{-2}n\text{Var}(X_1) = n^{-1}\text{Var}(X_1) = \frac{16}{n}(1 - C^{-1} - C^{-2}).$$

where the second equality follows from that the X_i s are independent.

Hence $\bar{X} \approx N(4/C, \frac{16}{n}(1 - C^{-1} - C^{-2}))$.

g)

`point.1` is the correct estimator of $E(\sqrt{X}) \approx n^{-1}\sum_i \sqrt{X_i}$. `point.2` may maybe be thought of as an estimate of $\sqrt{E(X)}$ ($\neq E(\sqrt{X})$ in general), but it will not be an unbiased estimator for finite n . `point.3` will always be equal to 1. `CI.2` is the correct CI.

Problem 2

a)

Since $N_1(t)$ is a homogeneous Poisson process, the times between events/until first event have exponential distributions with rate parameter $= \lambda$. The PDF is $f(t_1) = \lambda \exp(-\lambda t_1)$. The value of $E(T_1^2) = \text{Var}(T_1) + E(T_1)^2 = \lambda^{-2} + (\lambda^{-1})^2 = 2\lambda^{-2}$.

b)

Conditionally on $N_1(1) = 8$, these 8 event times will be independent and uniform(0,1)-distributed. The sought probability is therefore $\theta = P(N_1(0) = 0 | N_1(1) = 8) = P(U_1 > 0.1)P(U_2 > 0.1) \cdots P(U_8 > 0.1)$ for $U_1, \dots, U_8 \sim \text{iid } \text{Uniform}(0, 1)$. Hence $\theta = P(U_1 > 0.1)^8 = 0.9^8 \approx 0.4304672$.

Alternatively, this probability may be simulated something like

```
n <- 100000
Us <- matrix(runif(8*n), nrow=8)
n.before <- colSums(Us < 0.1)
print(paste0("estimated theta : ", mean(n.before == 0)))
```

```
## [1] "estimated theta : 0.43126"
```

c)

The distribution of a non-homogenous Poisson process at some fixed time t will be Poisson with mean $\int_0^t \lambda(s) ds$. In our case the mean is

$$\int_0^5 \lambda(s) ds = \int_0^5 1 + s/10 ds = 25/4 = 6.25.$$

I.e. in our case $N_2(5) \sim \text{Poisson}(6.25)$.

Problem 3

a)

The integrand of A may be factorized as

$$A = \int_0^\infty \underbrace{x^{3/2} \frac{1}{2}}_{g(x)} \underbrace{2 \exp(-2x)}_{f(x)} dx$$

which suggests the MC integration formula

$$\hat{A} = \frac{1}{2n} \sum_{i=1}^n X_i^{3/2}, \quad X_i \sim \text{iid } \text{Exponential}(1/2)$$

Sanity check (not required:)

```
n <- 1000000
x <- rexp(n, rate=2)
print(0.5*mean(x^1.5))
```

```
## [1] 0.2345881
```

b)

The integrand of B may be factorized as

$$B = \int_0^2 \exp(-2x)x^{3/2}dx = \int_0^\infty \underbrace{\mathbf{1}(x < 2)x^{3/2}}_{g(x)} \underbrace{\frac{1}{2}\exp(-2x)}_{f(x)} dx$$

where $\mathbf{1}(x < a)$ is 1 when $x < a$ and 0 otherwise. This suggests the MC integration formula

$$\hat{A} = \frac{1}{2n} \sum_{i=1}^n \mathbf{1}(X_i < 2) X_i^{3/2}, \quad X_i \sim \text{iid } \textit{Exponential}(1/2)$$

Sanity check (not required):

```
n <- 1000000
x <- rexp(n,rate=2)
print(0.5*mean((x<2.0)*x^1.5))

## [1] 0.1985984
```

c)

The uniform(-1,1)-distribution has density equal to 0.5 when $-1 < x < 1$, hence

$$C = \int_{-1}^1 \sqrt{1+x^3} dx = 2 \int_{-1}^1 \underbrace{\sqrt{1+x^3}}_{g(x)} \underbrace{\frac{1}{2}}_{f(x)} dx$$

which suggest the MC integration formula:

$$\hat{C} = \frac{2}{n} \sum_{i=1}^n \sqrt{1+U_i^3}, \quad U_i \sim \text{iid } \textit{Uniform}(-1,1).$$

similar reasoning for D :

$$\hat{D} = \frac{2}{n} \sum_{i=1}^n \sqrt{1+U_i^4}, \quad U_i \sim \text{iid } \textit{Uniform}(-1,1).$$

Sanity checks (not required):

```
u <- runif(1000000,min=-1,max=1)
print(2.0*mean(sqrt(1+u^3)))

## [1] 1.952323

print(2.0*mean(sqrt(1+u^4)))

## [1] 2.17855
```

d)

The gamma(1.5,0.5) distribution has density

$$f(x) = (1/2)^{-3/2} (\sqrt{\pi}/2)^{-1} x^{1.5-1} \exp(-2x) = 4\sqrt{\frac{2}{\pi}} \sqrt{x} \exp(-2x), x > 0.$$

Doing the usual importance sampling trick, we have that

$$A = \int_0^\infty x^{3/2} \exp(-2x) dx = \int_0^\infty \underbrace{\frac{x^{3/2} \exp(-2x)}{4\sqrt{\frac{2}{\pi}} \sqrt{x} \exp(-2x)}}_{g(x)} f(x) dx = \int_0^\infty \underbrace{\frac{x}{4\sqrt{\frac{2}{\pi}}}}_{g(x)} f(x) dx$$

The latter representation suggest the importance sampling estimator

$$\hat{A}_{is} = \frac{1}{n4\sqrt{\frac{2}{\pi}}} \sum_{i=1}^n X_i, \quad X_i \sim \text{Gamma}(1.5, 0.5)$$

The estimator will work fairly well as the importance weights do involve highly non-linear behavior.

Sanity check (not really required, but a plus if included): Due to the linearity of the importance weights, we can actually work out that

$$E(\hat{A}_{is}) = \frac{1}{4\sqrt{\frac{2}{\pi}}} E(X_i) = \frac{0.75}{4\sqrt{\frac{2}{\pi}}} \approx 0.2349964$$

i.e. equal to the hint.

Similar reasoning leads to

$$\hat{B}_{is} = \frac{1}{n4\sqrt{\frac{2}{\pi}}} \sum_{i=1}^n \mathbf{1}(X_i < 2) X_i, \quad X_i \sim \text{Gamma}(1.5, 0.5)$$

e)

Now, we consider proposals from $\text{Gamma}(2.5, 0.5)$ which has density

$$f(x) = (1/2)^{-5/2} \left(\frac{3\sqrt{\pi}}{4}\right)^{-1} x^{5/2-1} \exp(-2x) = \frac{16\sqrt{2}}{3\sqrt{\pi}} x^{3/2} \exp(-2x).$$

This leads to

$$A = \int_0^\infty \exp(-2x) x^{3/2} dx = \int_0^\infty \frac{\exp(-2x) x^{3/2}}{\frac{16\sqrt{2}}{3\sqrt{\pi}} x^{3/2} \exp(-2x)} f(x) dx = \frac{3\sqrt{\pi}}{16\sqrt{2}} \int f(x) dx$$

I.e. this results in the perfect (i.e. variance-free) importance sampling estimator

$$\tilde{A}_{is} = \frac{3\sqrt{\pi}}{16\sqrt{2}} \left(\frac{1}{n} \sum_{i=1}^n 1 \right) = A$$

and the corresponding estimate for B :

$$\tilde{A}_{is} = \frac{3\sqrt{\pi}}{16\sqrt{2}} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{1}_{X_i < 2} \right), \quad X_i \sim \text{iid } \text{Gamma}(2.5, 0.5).$$

We would expect also the latter to work really well as well.

f)

The basic antithetic associated with $U_i \sim \text{Uniform}(-1, 1)$ is simply $-U_i$. This suggest the following antithetic estimators based on n evaluations of the integrands:

$$\hat{C}_a = \frac{1}{2n/2} \sum_{i=1}^{n/2} \left(\sqrt{1+U_i^3} + \sqrt{1+(-U_i)^3} \right) = \frac{1}{n/2} \sum_{i=1}^{n/2} \left(\sqrt{1+U_i^3} + \sqrt{1-U_i^3} \right)$$

I.e. in this case with a monotonely increasing integrand, we would expect an improvement relative to the regular crude MC estimate for the same amount of computation

In the D -case, we get by similar reasoning the MC estimate

$$\hat{D}_a = \frac{1}{n/2} \sum_{i=1}^{n/2} \left(\sqrt{1+U_i^4} + \sqrt{1+(-U_i)^4} \right)$$

I.e. in this case, since the integrand is symmetric around $x = 0$, antithetic sampling actually decreases the effectivity. To see this, observe that

$$\hat{D}_a = \frac{1}{n/2} \sum_{i=1}^{n/2} \left(\sqrt{1+U_i^4} + \sqrt{1+U_i^4} \right) = 2 \frac{1}{n/2} \sum_{i=1}^{n/2} \sqrt{1+U_i^4}.$$

I.e. $\hat{D}_a$ is just wasting half of the integrand evaluations.

Sanity checks (not required)

```
nh <- 1000000
u <- runif(nh,min=-1,max=1)
print(mean(sqrt(1+u^3)+sqrt(1-u^3)))
```

```
## [1] 1.952753
```

```
print(mean(2*sqrt(1+u^4)))
```

```
## [1] 2.178805
```

Problem 4

a)

Bootstrap bias : mean of bootstrap samples minus estimator applied to original data set:

```
print(paste0("bias: ", 0.3841279 - 0.383732864871009))
```

```
## [1] "bias: 0.000395035128991028"
```

Bootstrap SD : simply the estimated standard deviation = square root of estimated variance:

```
print(paste0("SD: ", sqrt(0.001026572)))
```

```
## [1] "SD: 0.0320401622967175"
```

MSE: Variance + squared bias:

```
print(paste0("MSE: ", 0.001026572+(0.3841279 - 0.383732864871009)^2))
```

```
## [1] "MSE: 0.00102672805275314"
```

i.e. essentially the variance since the bias is so small

b)

Normal-based interval

```
dev <- 1.96*sqrt(0.001026572)
print(paste0("normal interval: ( ",
            round(0.383732864871009-dev,4)," , ",
            round(0.383732864871009+dev,4)," )"))

## [1] "normal interval: ( 0.3209 , 0.4465 )"
```

c)

Quantile-based interval : (0.327 , 0.449)

Comment: We see that the intervals are fairly similar, as it may be seen that the bootstrap samples are close being normal. Still the samples appear to be a tiny bit skewed to the right, which is the reason the quantile-based interval is shifted slightly to the right of the normal-based interval.

Problem 5

a)

This is a random walk Metropolis Hastings (RWMH) targeting a normal distribution with mean=2, SD=2. The proposal distribution is a normal centered at old state, with SD=3.

b)

We would expect the variance of the average of output from the RWMH to have a larger variance than a corresponding average based on iid samples. This is as the samples from the RWMH will be positively autocorrelated (dependent over time t) as can be seen from the plot (many instances of repeated values).